



SYNTHESIS AND CHARACTERIZATION OF IPOMONE

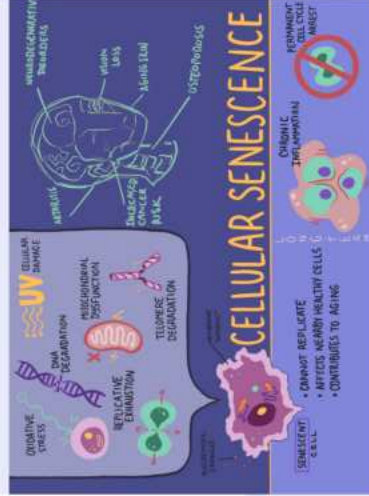


BACKGROUND

100,000

Number of people who die from complications related to **organismal aging** each day

SENESCENCE: AN ELEMENT OF ORGANISMAL AGING ASSOCIATED WITH PERMANENT CELL CYCLE ARREST AND MORPHOLOGICAL CHANGES



APOPTOSIS: A FORM OF PROGRAMMED CELL DEATH USED TO REMOVE DAMAGED CELLS FROM THE CELL CYCLE AND REGULATE SENESCENCE.

Figure 1) Diagram of causes and effects of senescence

Ipomone: a diterpenoid compound derived from morning glory Ipomoea Nil. Induces **apoptosis** by mediation of P-53 and Caspase-9 biochemical pathways.

PURPOSE

To synthesize and characterize the diterpenoid compound ipomone for use as a potential senolytic (anti-senescence agent)

VARIABLES

IV: Proportion of MTBE to Hexane used to separate ipomone from contaminants
DV: Amount and purity of ipomone obtained

PROCEDURE

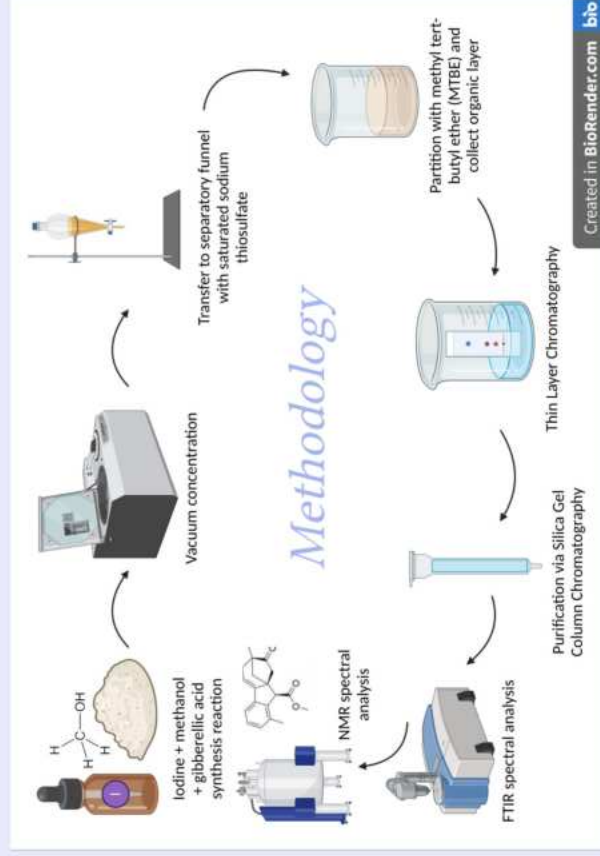


Figure 2) Diagram showing procedure for synthesis and identification of ipomone

RESULTS

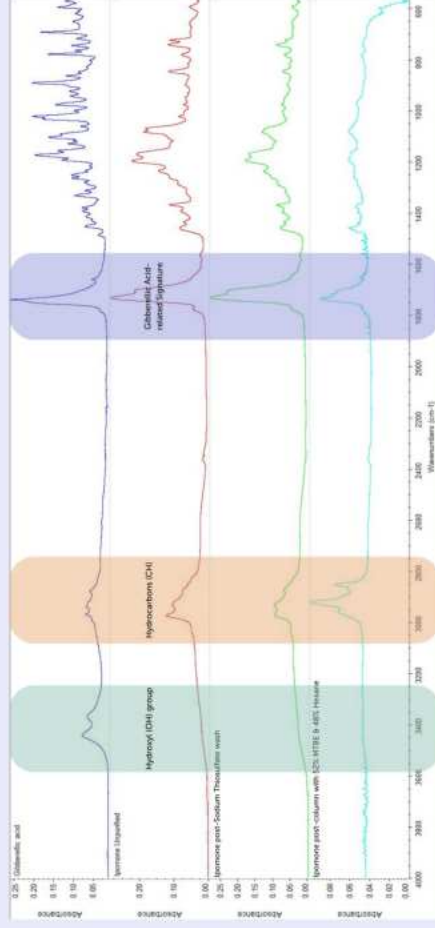
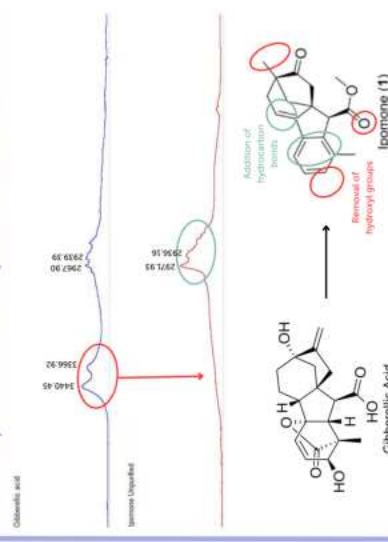


Figure 3) FTIR spectral analysis across synthesis phases: gibberellic acid, post-reaction mixture, post-sodium-thiosulfate wash, and post-column purification.

DISCUSSION

Due to the presence of contaminant peaks at wavenumbers 2800–3000 after the purification stage, it cannot be concluded that ipomone was successfully purified. However, due to the shift in peaks at 3200–3600 corresponding with a shifting hydroxyl group, it can be reasonably concluded that synthesis was achieved.



The addition of hydrocarbons and reduction in hydroxyl groups corresponds with the shift from gibberellic acid to the molecular structure of ipomone.

FUTURE WORK

Further work to remove contaminants must be undertaken before the ipomone compound obtained through this experiment can be applied. It will be assessed in a cellular biology context to examine its potential apoptotic and senolytic effects. Its potential as an anti-aging compound will be determined.

REFERENCES

Qiu, B., Chatterjee, E., Dey, B., Tripathi, N., Bhattacharya, N., Khatun, A., Ota, S. K., & Jain, S. K. (2021). Identification and evaluation of apoptosis-inducing activity of ipomoea nil from agricultural waste, unsaturated diterpenes. *Advanced Pharmaceutical Bulletin*, 9(2), 205–218. <https://doi.org/10.5717/apb.2019.024>

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